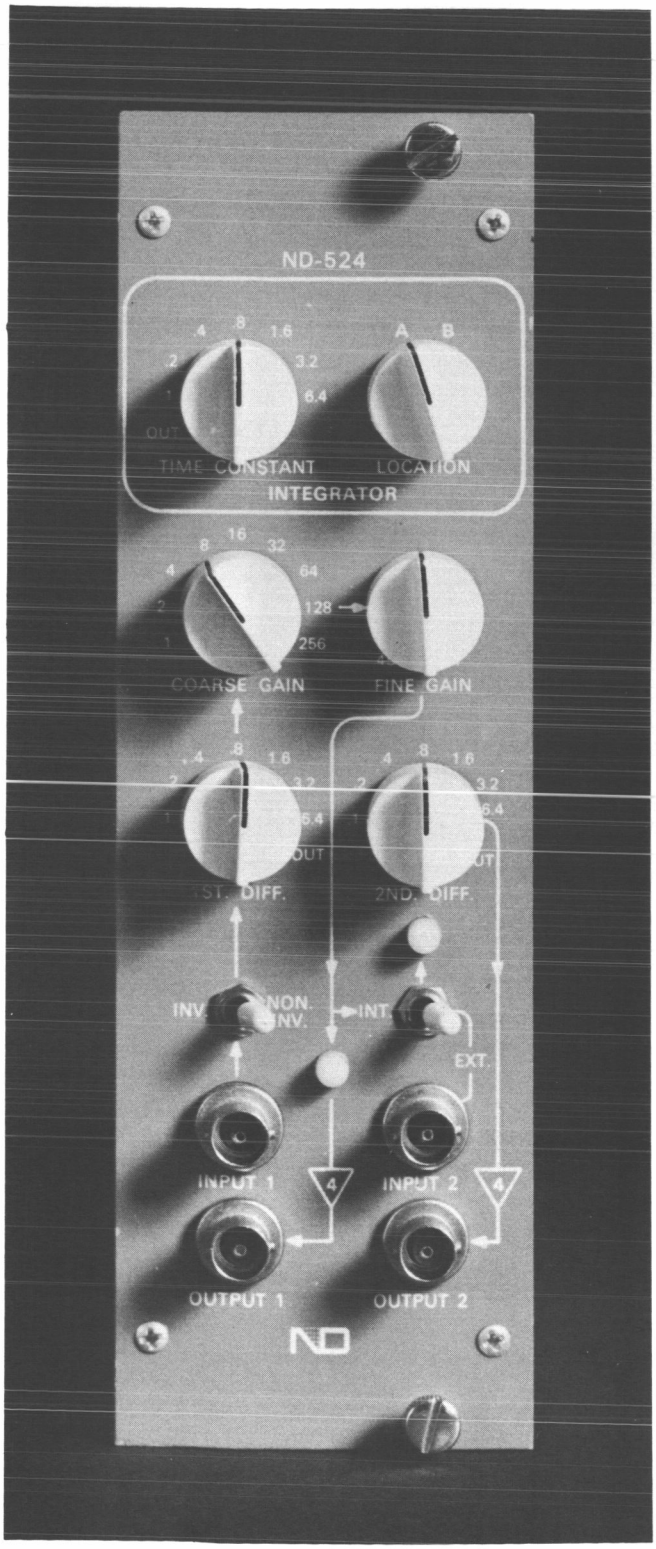
TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC.
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600



ND-524 LARC LINEAR AMPLIFIER

The ND-524 LARC is a high performance linear pulse amplifier specifically designed for high resolution nuclear pulse spectroscopy. It is especially well suited for application with solid state detectors where low noise, optimum pulse shaping, and excellent overload recovery are required. When used with an appropriate low noise preamplifier, the ND-524 provides the linear amplification and pulse shaping necessary for high resolution multichannel analysis with analog-to-digital converters such as used in the Nuclear Data Series 2200 and 3300 systems. R.C. pulse shaping is provided by two independently adjustable differentiators, and an independently adjustable double integrator. The first differentiator is pole-zero compensated to minimize undershoot effects, and the integrator, which utilizes both passive and active elements, provides a more nearly gaussian response to improve signal-to-noise ratio.

Sectionalized construction facilitates insertion of discriminators, linear gates, timing pick-offs or other similar instrumentation. The input section, which has a maximum gain of 256, contains the first differentiator and the amplifier gain controls. Two independent output sections each have a fixed gain of four. The second differentiator is located within one of the two output sections, and the integrator location can be switch selected to precede either output section. If desired, the two output sections can be cascaded to provide a maximum amplifier gain of 4096. The ND-524 LARC is packaged in a standard AEC double width module, so that up to six units may be housed in a single AEC bin. All inputs and outputs appear on both the front and rear panels to simplify interconnection. A D. C. power output for operation of the companion ND-525 HRP high resolution preamplifier is also provided.

SPECIFICATIONS

Input 1:

Two BNC connectors (one front, one rear). Input accepts positive or negative low level signals (D. C. level not to exceed 15 volts). Input impedance; 1000 ohms ±1% for positive or negative signal.

Input 2:

Two BNC connectors (one front, one rear). Input accepts positive or negative 0-10 volt signals (D. C. level not to exceed 15 volts). This input is coupled to a final amplification section which has a fixed gain of X4. Input impedance; 1000 ohms ±1% for positive or negative signals.

Polarity:

For signals applied to INPUT 1; output is the same polarity as input (non-inverted), or output is opposite polarity as input (inverted).

Gain Capability:

Gain of 1024 for output pulses with one integration and two differentiations of equal time constant. Gain is continuously adjustable over a range of 4-1024. The two output sections can be cascaded to provide a gain of 4096.

Gain Stability:

Less than 0.005% per degree C and 0.01% per 10% change in line voltage.

Pulse Shaping:

Double integration and two differentiations, with time constants independently variable in 2:1 steps over a range from 0.1 to 6.4 microseconds. In addition, it is possible to disable each differentiation and the double integration individually, so that with all three disabled, the amplifier has an output rise time of 80 nanoseconds and fall time of 10 milliseconds, for a step function input.

Noise:

Less than 8 microvolts, referred to input, for a singly differentiated — double* integrated pulse with 1.6 microsecond time constant at all gains from 64-1024.

Integral linearity:

Less than 0.1% over the output range of 0.1 to 10 volts open circuited, or 0.05 to 5 volts into 100 ohms, either polarity.

Differential Linearity:

Less than 0.4% over the output range of 0.1 to 10 volts open circuited, or 0.05 to 5 volts into 100 ohms, either polarity.

Crossover Walk:

Not more than ±2 manoseconds for doubly differentiated pulses over full linear output range and not more than ±1 manosecond over full gain control range.

Overload Performance:

Double differentiated — double* integrated: Recovery to 1% of rated output in 2 non-overloaded pulse widths for an overload of 500 times for all time constants.

Single differentiated — double* integrated: An overload of 100 times produces an undershoot not greater than 5% of full scale for time constants up to 1.6 microseconds.

*Double Integrator: consists of two ganged low pass filters with identical response, followed by a non-inverting amplifier with a gain which compensates for filter attenuation.

Maximum Count Rate:

No limitation on count rate.

Pole Zero Compensated:

The first differentiator is compensated for undershoot effects at all selected time constants. Optimum input decay time for pole zero compensation is 55 microseconds ±5%.

Output 1:

Two BNC connectors (one front, one rear) provide a 10 volt, open circuit, output signal. Impedance is 100 ohms ±1%, short circuit protected. The two outputs have a common source impedance of 0.2 ohms.

Output 2:

Two BNC connectors (one front, one rear) provide a 10 volt, open circuit, output signal. Impedance is 100 ohms ±1%, short circuit protected. The two outputs have a common source impedance of 0.2 ohms.

Connectors:

All signal connectors BNC, duplicated on front and rear panels. Preamplifier D. C. power connector Amphenol, type 126-218, located on rear panel.

Power Requirements:

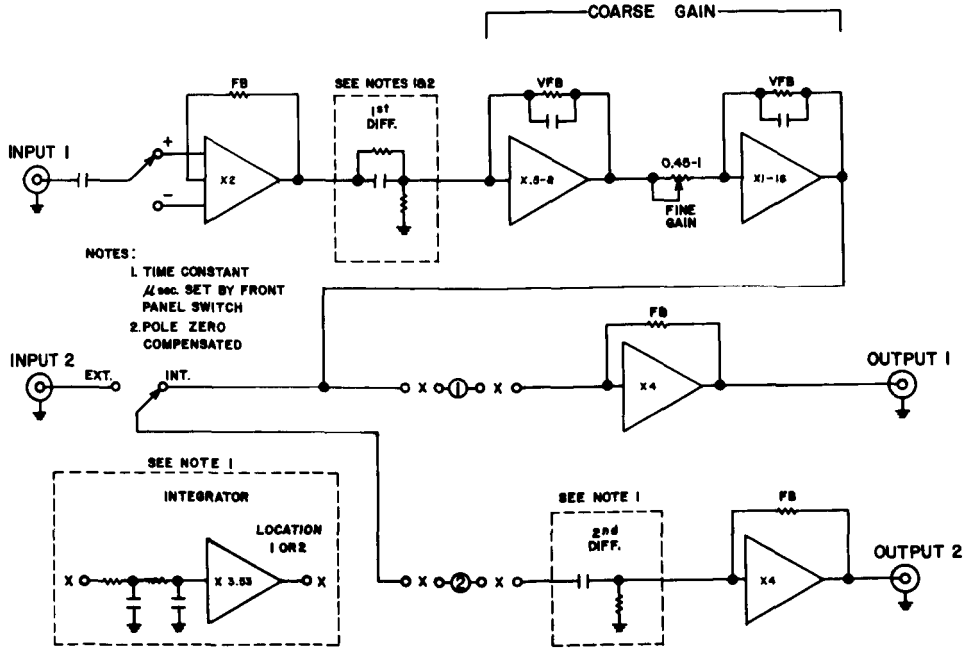
+24 volts at 110 ma.
−24 volts at 110 ma.
+12 volts at 60 ma.

Size:

8.71" high x 2.68" wide x 9.7" deep (Standard AEC Double Width).

Weight:

Approximately 2 pounds and 8 ounces.



BLOCK DIAGRAM OF ND-524 LARC

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